

Ahmed Farahat

Mechatronics Engineer

Contact Information

Address 1: Hadayek Al-Qubba, Cairo, Egypt

Phone: [+201119027639](tel:+201119027639)

LinkedIn: [linkedin.com/in/ahmed-farahat-87038320b](https://www.linkedin.com/in/ahmed-farahat-87038320b)

Address 2: 6th of October, Egypt

Email: ahmedfarahat1177@gmail.com

GitHub: github.com/Baby-Madara

Professional Summary

Mechatronics Engineer focusing on **Robotics, AI-driven Control Systems, and Software Engineering**. Experienced in **Embedded Systems, ROS & ROS2, SolidWorks, MATLAB, and Web Development**. Driven to contribute creative solutions to advanced engineering challenges.

Education

B.Sc. Mechatronics Systems Engineering

MSA University, (October 2019 – June 2024)

GPA: 3.77 (Ranked 5th)

General Secondary Education

El-Nokrashy Secondary School, (July 2019)

Grade: 92.4%

STEM October School

(September 2016 – July 2018)

Professional Experience

GAT Solutions

Mechatronics Engineer Intern, (January 2026)

Developing software applications to support mechanical engineering workflows.

Little Engineer Academy

Part-Time Robotics Instructor, (October 2025 – January 2026)

Teaching robotics concepts to 8-20 years old students through hands-on learning based on *Arduino*.

Arax Pharma

Customer Service Agent, (June 2025 – August 2025)

Provided support for customer through *social media chat* platforms.

FiveM Group

Full-Stack Robotics Engineer, (February 2025 – June 2025)

Mechanical and control design of a custom delta robot (project-based contract)

ARIA Technologies – Summer Internship Program

Mechanical and Automations Departments, (July 2024 – September 2024)

Contributed to the development of *ARIA P60 Robot Arm*.

Elmarakby Steel – Summer Internship Program

Hydraulics and Heavy Equipment Departments, (August 2022)

Gained theoretical and hands-on experience with hydraulic systems and heavy machinery.

AMIT Learning – Embedded Systems Diploma

Graduation Project: Smart Home System (Grade: 95%), (July 2022 – November 2022)

Interfacing mainly based on AVR microcontrollers with introduction to ARM architecture.

Arab Contractors

Engine Mechanics Trainee, (September 2021)

Practical Training in Engine Mechanics

Technical Skills

Programming Languages: C++, C, Python, JavaScript

Web Development: Frontend (HTML, CSS, JS), MERN (MongoDB, Express, React, Node.JS), Network Protocols (HTTP, WebSockets), GitHub Pages

Embedded Systems: Embedded C & C++, RTOS, Serial Communication Protocols (UART, I2C, SPI), GPIOs

SBCs & Microcontrollers: Raspberry Pi, NVIDIA Jetson Nano, Arduino, ESP32, ATmega32

Linux OS Ubuntu

Development Tools: Version Control (Git & GitHub), Visual Studio Code, PlatformIO

Robotics & Control Tools: MATLAB & Simulink, ROS & ROS2, PyBullet Physics Simulator

Artificial Intelligence: Machine Learning, Reinforcement Learning (tools: Python Stable Baseline3: PPO)

Industrial Automation: Beckhoff IPC, PLC Programming (Ladder Diagram, Structured Text, Continuous Flow Chart)

CAD & Simulation Tools: SolidWorks (Motion Analysis, Topology Optimization, Sheet Metal Design), Autodesk Inventor, Autodesk Fusion 360, ANSYS, CNC, 3D Printing

PCB Design Basics

Graphic Design: Adobe Illustrator, Photoshop, Blender

Engineering Projects

Graduation Project: “Development of an Autonomous Quadruped Robot”

- 12 DoF Legged Robot optimized with **motion analysis** and **topology optimization** approaches
- Developed using **ROS2** & **MATLAB**. Simulated on **PyBullet**
- Final implementation on **ESP32s (reduce costs by 14k EGP)**
- **OTA** feature (uploading firmware wirelessly, about x2 upload speed), enabling conversion to be an IoT project
- Operated from web browsers over Wi-Fi (range in open areas: ~50 meters)
- RTOS-Based Firmware
- Two reinforcement learning controllers & Modified Locomotion Pattern Controller

Other Projects:

Portfolio Website

- One-page website – Made by ReactJS – Hosted on GitHub Pages – Responsive, interactive, light/dark themes
- Link: [Portfolio - Ahmed Farahat](#)

Scientific Calculator

- Modelled as one-page website, using JavaScript – Hosted on GitHub Pages – Link: [Scientific Calculator](#)

Field-Oriented Control for BLDC Motor with Three Different Controllers: PID, Fuzzy Logic, and Neural Network

Reinforcement Learning Agents for VizDoom Game (4 game modes, PPO Algorithm)

Smart Home System

- Accessed over Bluetooth – Multi-user, each with password and different privileges – Data storage on EEPROM
- Link: [here](#)

EMG-based Prosthetic Arm with Force Feedback

- Finalist at Innovation Catalyst 2022
- Features 2 modes (automatic/manual)
- Involved custom mechanical design, OOP programming, PCB design and fabrication over two iterations

CNC Foam Cutting Machine

Self-Balancing Robot

Sensors: gyroscope & accelerometer – Controller: PID

Web-Browser-Controlled Airplane Disinfecting Robot

Controlled from web browsers – Broadcasts video streaming to user over Wi-Fi – System implemented on ESP32 cam.

3D Printer (Homemade)

MMR Gearbox (Mechanical Motion Rectifier)

Rectifies output rotation directions in one direction.

Mini-Recording Piano using Arduino

Enables addition of tens of piano keys (signal is analog rather than DIO pin for each key) – Simulation Link: [here](#)

Student Activities & Competitions

Technical Head Team, IEEE(MSA)

Technical Head, **(November 2022 – November 2023)**

Innovation Catalyst (Competition)

X-Bionics Team – **Finalist, (April 2022 – September 2022)**

3-Days Arduino Workshop, IEEE(MSA)

Session Presenter, **(April 2022)**

Technical Head Team, IEEE(MSA)

Technical Vice Head, **(November 2020 – July 2022)**

Motivated team members and engaged in different tasks

Additional Information

Military Status: Exempted